

- 3B. In a normal distribution 7% of the items are under 35 and 89% are under 63. Find the mean and standard deviation.
- 3C. If the random variable 'K' is uniformly distributed over $[0, 5]$, what is the probability that the roots of the equation $4x^2 + 4xK + K + 2 = 0$ are real ? **(4 + 3 + 3)**
- 4A. Two independent observations X_1 & X_2 are made on a random variable X with density function $f(x, \theta) = \frac{1}{\theta} e^{-x/\theta}$ for $0 < x < \infty$ and $\theta > 0$. To test $H_0 : \theta = 2$ against $H_1 : \theta = 4$ it is decided to reject H_0 if $x_1 + x_2 \geq 9.5$ and accept otherwise. Obtain the significance level and power of the test.
- 4B. Compute an approximate probability that mean of a random sample of size 15 from a distribution having pdf $f(x) = \begin{cases} 3x^2, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$ is between $\frac{3}{5}$ and $\frac{4}{5}$.
- 4C. Suppose that the continuous random variable X has pdf $f(x) = \frac{1}{2} e^{-|x|}$, $-\infty < x < \infty$. Find mgf of X and hence find $E(X)$ and $V(X)$. **(4 + 3 + 3)**
- 5A. Let $(X_1, X_2, \dots, X_n)$ be a random sample of size n from a distribution having pdf $f(x, \theta) = \begin{cases} \theta x^{\theta-1}, & 0 < x < 1, 0 < \theta < \infty \\ 0, & \text{elsewhere} \end{cases}$ Find M. L. E. for θ .
- 5B. Show that for a normal distribution with mean μ and variance σ^2 , $E[(X - \mu)^{2n}] = 1.3.5 \dots (2n-1) \sigma^{2n}$.
- 5C. Suppose that the two dimensional random variable (X, Y) is uniformly distributed over the triangular region $R = \{(x, y) / 0 < x < y < 1\}$. Find $\text{COV}(X, Y)$. **(4 + 3 + 3)**
- 6A. A fair coin is tossed till first head appears. Let X denote the number of tosses. Find $E(X)$ and $V(X)$.
- 6B. A binary communication channel carries data as one of two types of signals denoted by 0 and 1. Due to noise a transmitted 0 is received as 1 with probability 0.6 and 1 is received as 0 with probability 0.09. Probability of transmitting 0 is 0.45. If a signal is sent determine
(i) probability that 1 is received.
(ii) probability that 0 was transmitted given that 0 was received.
- 6C. Show that the sample mean $\bar{X}$ is both unbiased and consistent estimator for the population mean. **(4 + 3 + 3)**
